Midway Milestone Report For ECE285

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Abstract

This project aims to build **SolarInvestAgent**, an intelligent decision-support system that optimizes the sizing of residential Photovoltaic (PV) systems and Battery Energy Storage Systems (BESS). Unlike standard Large Language Models (LLMs) that rely on heuristics, this agent will utilize a rigorous Mixed-Integer Nonlinear Programming (MINLP) tool. Drawing on the methodology established by Zhou et al. (2018), the agent will explicitly account for tariff structures (Stepwise vs. Time-of-Use), seasonal weather uncertainty, and battery degradation constraints to generate mathematically optimal investment advice. The agent will leverage five years of historical household consumption data to model demand variability and determine the optimal PV technology, system capacity, and installation size that minimizes total lifecycle cost under tariff, weather, and degradation constraints.

1 Introduction

Buildings account for approximately 40% of global end-use energy consumption, making residential energy efficiency a critical component of carbon reduction strategies. However, homeowners face complex techno-economic decisions when sizing photovoltaic (PV) and battery energy storage systems (BESS), particularly under varying tariff structures and uncertain weather conditions.

- **Objective A:** Develop a baseline Large Language Model (Track A) that provides heuristic, chatbot-style solar investment recommendations.
- **Objective B:** Develop an agentic framework (Track B) capable of invoking a Python-based Mixed-Integer Nonlinear Programming (MINLP) optimization tool adapted from Zhou et al. (2018), with primary focus on Time-of-Use (TOU) tariff structures, seasonal weather uncertainty, and battery degradation constraints.
- **Objective C:** Compare both tracks to demonstrate that the agentic optimization-based approach yields superior financial accuracy and consistency under Time-of-Use (TOU) pricing, with Stepwise Power Tariff (SPT) considered as an optional extension if time permits.

2 Related Work

Optimal sizing of residential photovoltaic (PV) and battery energy storage systems (BESS) has been widely studied using mixed-integer and stochastic optimization frameworks embedded within home energy management systems. Early foundational work by Zhou et al. Zhou et al. [2018] formulated a bi-level mixed-integer nonlinear program (MINLP) to jointly determine PV and BESS capacities under stepwise power tariffs (SPT), time-of-use (TOU), and real-time pricing (RTP). Their results demonstrated that tariff structure significantly influences investment decisions: PV-only systems tend to be optimal under stepwise tariffs, whereas combined PV–BESS configurations

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become economically attractive under TOU and RTP when uncertainty and subsidy structures are incorporated.

Subsequent research has refined and extended this formulation using mixed-integer linear programming (MILP), stochastic programming, and scenario-based methods to improve tractability and realism. These studies incorporate battery degradation modeling, flat versus TOU tariff comparisons, demand response participation, and integration of additional flexible loads such as electric vehicles Afzal et al. [2018], Hassan et al. [2021], Hanna et al. [2017]. More recent stochastic frameworks explicitly model joint uncertainty in PV generation, household demand, and electricity prices while penalizing deep battery cycling to capture lifetime degradation effects Wang et al. [2024]. Collectively, this literature establishes that under TOU pricing structures, optimal battery sizing critically depends on price spread between peak and off-peak periods, load coincidence with PV production, and battery degradation economics.

Despite their rigor, these optimization frameworks are typically implemented as standalone engineering tools requiring domain expertise in mathematical programming, parameter calibration, and scenario design. As a result, they are rarely directly accessible to individual homeowners or small installers Zhou et al. [2018], Afzal et al. [2018].

In parallel, emerging industry platforms and conceptual work on agentic AI systems demonstrate conversational "solar assistants" capable of estimating return on investment (ROI), recommending system sizes, and answering policy-related questions through large language model (LLM) interfaces Digiqt [2025], OnixNet [2025]. However, these systems generally rely on rule-based heuristics, simplified payback calculations, or static lookup tables rather than tightly coupled stochastic optimization models.

SolarInvestAgent builds on the stochastic PV–BESS sizing literature—particularly TOU-driven formulations—while introducing a fundamentally different system architecture. Instead of presenting optimization as a backend engineering tool, we embed a Zhou-style stochastic sizing model within an LLM-driven agentic framework that orchestrates data collection, scenario generation, solver execution, and explanation synthesis. This integration enables mathematically grounded investment recommendations to be delivered through a natural-language interface, bridging the gap between rigorous stochastic optimization and accessible decision support for residential users.

Unlike prior optimization-only studies, our contribution lies not in a novel tariff formulation per se, but in the architectural coupling of stochastic energy system design with agentic reasoning and conversational interaction, evaluated under realistic TOU pricing structures and uncertainty.

3 Methodology

Our methodology bridges rigorous mathematical optimization with an accessible Large Language Model (LLM) agent wrapper. To evaluate the feasibility of residential Solar PV and Battery Energy Storage Systems (BESS), we constructed a deterministic and stochastic pipeline spanning data acquisition, physical system modeling, and financial optimization.

3.1 Problem Formulation

The core objective of **SolarInvestAgent** is to minimize the total lifecycle cost (or equivalently, maximize the Net Present Value) of meeting a household's electricity demand over a 10-year horizon. Drawing from the MINLP formulation of Zhou et al. Zhou et al. [2018], our objective function explicitly evaluates capital expenditures, operational maintenance, and the dynamic grid exchange costs under San Diego Gas & Electric's (SDG&E) Time-of-Use (TOU) tariff structure:

$$\min\ C_{\text{total}} = C_{\text{PV}} + C_{\text{BESS}} + C_{\text{O\&M}} + \sum_{y=1}^{10}\sum_{t=1}^{8760}\left[P_{\text{grid}}^{\text{buy}}(t)\cdot\lambda_{\text{TOU}}^{\text{buy}}(t) - P_{\text{grid}}^{\text{sell}}(t)\cdot\lambda_{\text{NEM}}^{\text{sell}}(t)\right] \quad (1)$$

where C_{PV} and C_{BESS} represent the amortized capital costs of the solar and storage systems, respectively. $\lambda_{\text{TOU}}^{\text{buy}}(t)$ is the time-varying TOU electricity purchase rate (with extreme peak pricing from 4 PM–9 PM under SDG&E's EV-TOU schedule), and $\lambda_{\text{NEM}}^{\text{sell}}(t)$ is the net energy metering export compensation rate under California's NEM 3.0 Net Billing framework. The decision variables

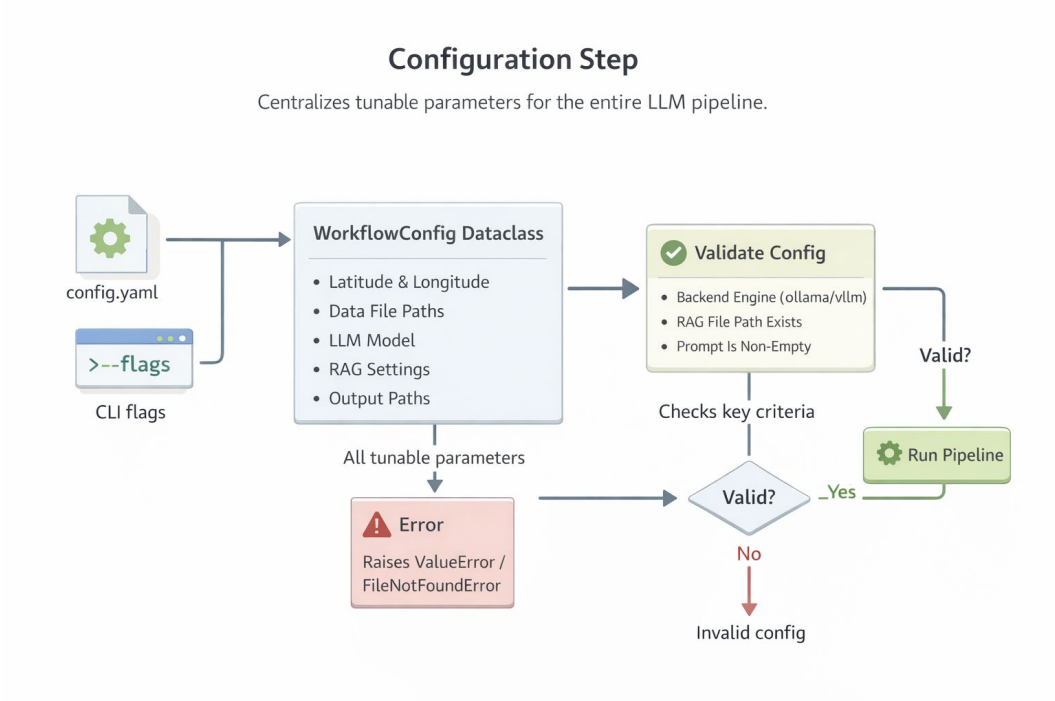

Figure 1: Centralized configuration and validation layer controlling model parameters, file paths, and backend selection prior to pipeline execution.

are the installed PV capacity S_{PV} (kW_p) and the BESS rated energy capacity E_{BESS} (kWh). The annual O&M cost is modeled dynamically as $C_{\text{O\&M}} = N_{\text{panels}} \cdot \$15/\text{panel/visit} \cdot 2$ visits/year $+$ \$250/year (inspection), yielding approximately \$350–\$500 annually for a standard San Diego residential system.

To ensure physical viability and safe operation of the BESS, the system is subjected to the following hourly constraints:

- **Power Balance (at every hour t):**

$$P_{\text{load}}(t) = P_{\text{grid}}^{\text{buy}}(t) - P_{\text{grid}}^{\text{sell}}(t) + P_{\text{PV}}(t) + P_{\text{dch}}(t) - P_{\text{ch}}(t) \tag{2}$$

- **State-of-Charge (SoC) Bounds:** To prevent deep cycling and extend battery lifetime:

$$0.2 \cdot E_{\text{BESS}} \leq \text{SoC}(t) \leq 0.8 \cdot E_{\text{BESS}} \tag{3}$$

- **SoC Dynamics:**

$$\text{SoC}(t) = \text{SoC}(t-1) + \eta_{\text{ch}} P_{\text{ch}}(t) - \frac{P_{\text{dch}}(t)}{\eta_{\text{dch}}} \tag{4}$$

- **Mutual Exclusivity of Charging/Discharging:**

$$x_{\text{ch}}(t) + x_{\text{dch}}(t) \leq 1, \quad x_{\text{ch}}, x_{\text{dch}} \in \{0, 1\} \tag{5}$$

The binary variables $x_{\text{ch}}(t)$ and $x_{\text{dch}}(t)$ are what render this a Mixed-Integer Nonlinear Program (MINLP), as they interact with continuous power variables in the SoC dynamics.

3.2 Stochastic Scenario Generation

Rather than optimizing over a single "average" day—which would systematically underestimate the cost of demand peaks and irradiance troughs—we employ a stochastic approach (SA) following the scenario-weighting methodology of Zhou et al. We generate 12 representative scenarios by crossing four seasons (Spring, Summer, Autumn, Winter) with three weather conditions (Sunny,

Partly Cloudy, Overcast/Rainy), each assigned an empirical probability weight π_s derived from five years of historical Open-Meteo cloud cover data for the San Diego coordinates (Latitude 32.7157°N, Longitude 117.1611°W). The stochastic objective is:

$$\min \sum_{s=1}^{12} \pi_s \cdot C_{\text{total},s}(S_{\text{PV}}, E_{\text{BESS}}) \tag{6}$$

This formulation ensures that the recommended system sizing is robust across realistic seasonal and meteorological variability, rather than being over-optimized for best-case solar conditions.

3.3 Data Extraction and Synthetic Generation Pipeline

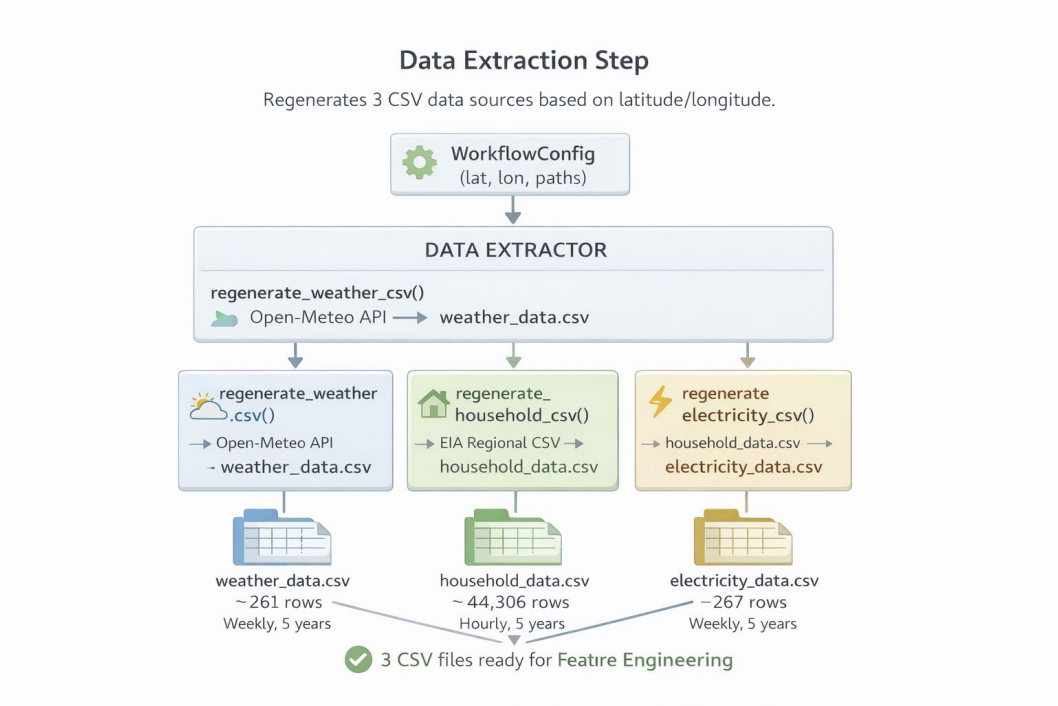

Figure 2: Data extraction pipeline architecture. The system regenerates three core datasets (weather, household load, electricity tariff data) from latitude/longitude inputs, producing structured CSV outputs used for feature engineering and optimization.

To parameterize the optimization model with highly granular, localized data specific to San Diego, we developed two automated Python extraction pipelines:

3.3.1 Weather and Irradiance Data (`weather_extraction.py`)

We interface with the Open-Meteo Historical Weather API to retrieve five years of historical weather records (from February 2019 to February 2024) anchored to the downtown San Diego coordinates. The pipeline extracts, at hourly resolution: (i) ambient temperature (°C), (ii) cloud cover (%), and (iii) shortwave radiation (W/m^2, the primary driver of PV generation). For each variable, we compute daily maximum, minimum, and mean statistics, which are then aggregated into weekly representative profiles. This weekly aggregation reduces noise while preserving seasonally meaningful variation. The output is a clean CSV (`weather_data.csv`) with 261 weekly rows and 10 feature columns:

```
week_number, weekly_{max,min,avg}_temperature,
weekly_{max,min,avg}_irradiance, weekly_{max,min,avg}_cloud_cover
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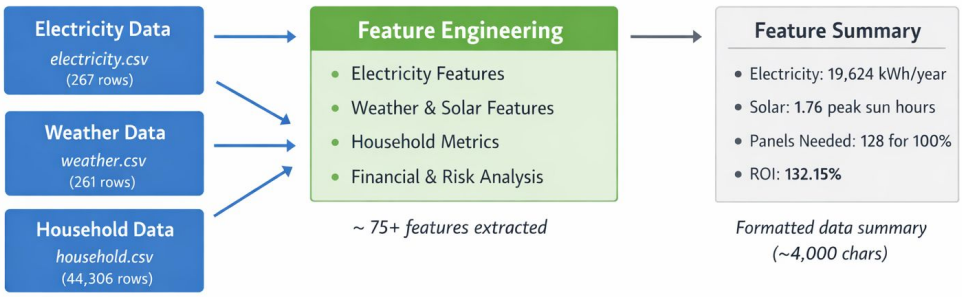

Figure 3: Feature engineering stage aggregating weather, load, and tariff datasets into 75+ structured features for analysis.

3.3.2 Household Load Generation (`household_extraction_per_house.py`)

Recognizing that aggregate regional load profiles from grid operators fail to capture the heterogeneity of individual residential demand, we developed a synthetic downscaling generator. The pipeline begins with hourly MW-level load data for the San Diego Gas & Electric (SDGE) sub-balancing authority, downloaded from the U.S. Energy Information Administration (EIA) API (`EIA_SD_Load_Download.py`) covering January 2021 to February 2026. This regional signal is downscaled to an individual household using nine deterministic variability factors grouped into three categories:

1. **Geographic Scalars:** A longitude-based coastal/inland microclimate factor (homes closer to the ocean exhibit lower cooling loads due to the marine layer); an elevation proxy modifying heating degree-days; and a north-south temperature gradient scalar.
2. **Household Structural Characteristics:** Conditioned square footage (load scales approximately linearly with floor area above a base load threshold); economic vintage of the home (older construction has lower insulation R-values, increasing HVAC load); and a neighborhood density factor (urban heat island effect for dense areas).
3. **Stochastic Time-Series Modifiers:** A probabilistic Electric Vehicle (EV) charging schedule that injects 3–7 kW nighttime load spikes (10 PM–6 AM) with a user-specified daily charging probability; and an existing-solar daylight suppression curve that attenuates midday grid consumption proportional to assumed pre-existing PV penetration in the neighborhood.

The combined output is a synthetic 8760-hour annual household load profile (kW) that is simultaneously anchored to measured regional grid behavior and enriched with realistic individual-level variance—precisely the kind of "messy," non-smooth signal that stresses an optimizer and produces more conservative, real-world-ready system sizing recommendations.

3.4 Hierarchical Solar Production Model

To map weather data onto hourly PV power output, we developed a hierarchical electrical model that scales from a single photovoltaic cell to the complete installed array. Let P_c, V_c, and I_c denote

the rated maximum power, open-circuit voltage, and short-circuit current of a single solar cell under Standard Test Conditions (STC: $G_{\text{ref}} = 1000$ W/m^2, $T_{\text{ref}} = 25$°C). For an array configured with n_s cells in series per module, n_p cell strings in parallel per module, N_s modules in series per string, and N_p strings in parallel, the idealized STC array power is:

$$P_{\text{array}}^{\text{STC}} = N_s \cdot N_p \cdot n_s \cdot n_p \cdot P_c \tag{7}$$

Incorporating our dynamically extracted irradiance and temperature data, the actual hourly generation $P_{\text{PV}}(t)$ is modeled as:

$$P_{\text{PV}}(t) = P_{\text{array}}^{\text{STC}} \cdot \frac{G(t)}{G_{\text{ref}}} \cdot [1 + \gamma\,(T_{\text{cell}}(t) - T_{\text{ref}})] \cdot PR \cdot (1-d)^y \tag{8}$$

where $G(t)$ is the hourly extracted shortwave irradiance (W/m^2), γ is the temperature coefficient of power (typically -0.004 %/°C for monocrystalline silicon), $T_{\text{cell}}(t)$ is the estimated cell temperature derived from ambient temperature and irradiance via the NOCT model, PR is the system Performance Ratio accounting for wiring losses, inverter efficiency, soiling, and mismatch (typically 0.75–0.85), d is the annual degradation rate (typically 0.5%/year for premium panels), and y is the year of operation. This formulation is parameterized by a panel specification dictionary, allowing us to seamlessly swap between different manufacturers' datasheets—a key feature for the cost comparison module.

3.5 Tariff Modeling: Time-of-Use (TOU) Structure

A pivotal architectural decision was the shift from Stepwise Power Tariffs (SPT)—the primary structure analyzed in the Zhou et al. reference paper—to Time-of-Use (TOU) pricing, which is the dominant and legally mandated residential tariff structure in the SDG&E service territory. Under SDG&E's EV-TOU rate schedule, electricity prices vary by time-of-day and season:

Table 1: SDG&E TOU-DR Rate Schedule by Segment, 2021–2025 (total \$/kWh inclusive of all charges). Sourced from publicly available SDG&E regulatory PDF tariff schedules, manually extracted and validated. From October 2025 onward, the daily fixed charge structure transitioned from a Minimum Daily Bill to a Base Services Charge (\$0.793/day), with energy-only rates shown.

Year	**Dates**	**Summer (\$/kWh)**			**Winter (\$/kWh)**	
		On-Pk	**Off-Pk**	**S-Off-Pk**	**On-Pk**	**Off-Pk**
2021	Jun–Oct 2021	0.490	0.428	0.368	0.413	0.404
2022	Jun–Dec 2022	0.565	0.494	0.416	0.543	0.481
2023	Jan–Dec 2023	0.666	0.569	0.465	0.636	0.552
2024	Mar–Dec 2024	0.570	0.502	0.428	0.553	0.494
2025 (a)	Feb–Sep 2025	0.599	0.528	0.450	0.582	0.519
2025 (b)†	Oct–Dec 2025	0.535	0.463	0.386	0.517	0.455

TOU periods (weekdays): On-Peak 4–9 p.m.; Off-Peak 6 a.m.–4 p.m. & 9 p.m.–midnight; Super Off-Peak midnight–6 a.m. (plus 10 a.m.–2 p.m. in March–April). Summer: Jun 1–Oct 31; Winter: Nov 1–May 31.

† Oct–Dec 2025 uses the new Base Services Charge structure (\$0.793/day fixed); energy-only rates shown.

Additionally, under California's NEM 3.0 Net Billing Tariff (effective April 2023), the export compensation rate for rooftop solar has been reduced dramatically compared to previous NEM 2.0 rates—to approximately \$0.05–\$0.08/kWh on average—fundamentally changing the economics of PV-only systems and making BESS a near-necessity for financially viable residential solar installations in San Diego.

3.6 Hardware Cost Aggregation

We surveyed current market pricing from nine solar panel brands and leading battery storage distributors active in the San Diego market to populate our cost model EnergySage [2025]. Panel costs

are specified as \$2.85–\$3.75/W_p installed (inclusive of racking, inverter, and installation labor), and BESS unit costs range from \$7,000–\$18,000 per unit (covering 10–13.5 kWh of usable capacity, inclusive of BMS and installation) National Renewable Energy Laboratory [2024a]. We collected datasheets and pricing from REC Group, JA Solar, Trina Solar, Canadian Solar, Silfab, Jinko Solar, LONGi Solar, Maxeon Solar, and Aiko Solar, spanning an installed cost range of \$2.85–\$3.75/W_p at efficiencies of 21.8%–24.3% (Table 2).

Table 2: Solar Panel Brand Costs and Efficiencies for San Diego (2025) EnergySage [2025]

Brand (Model)	**Installed Cost**	**Efficiency**	**Best For**
REC Group (Alpha Pure)	\$2.85/W_p	22.6%	Best overall value
JA Solar (DeepBlue)	\$2.90/W_p	–	Competitive budget option
Trina Solar (Vertex S)	\$2.90/W_p	21.8%	Cost-sensitive installations
Canadian Solar (TOPHiKu7)	\$3.10/W_p	22.5%	High power output
Silfab Solar (Prime)	\$3.15/W_p	22.1%	North American manufacturing
Jinko Solar (Tiger Neo)	\$3.20/W_p	23.8%	Premium performance
LONGi Solar (Hi-MO 6)	\$3.35/W_p	23.3%	Innovation leader
Maxeon Solar (Maxeon 7)	\$3.50/W_p	22.8%	40-year warranty
Aiko Solar (Neostar 2P)	\$3.75/W_p	24.3%	Highest efficiency available

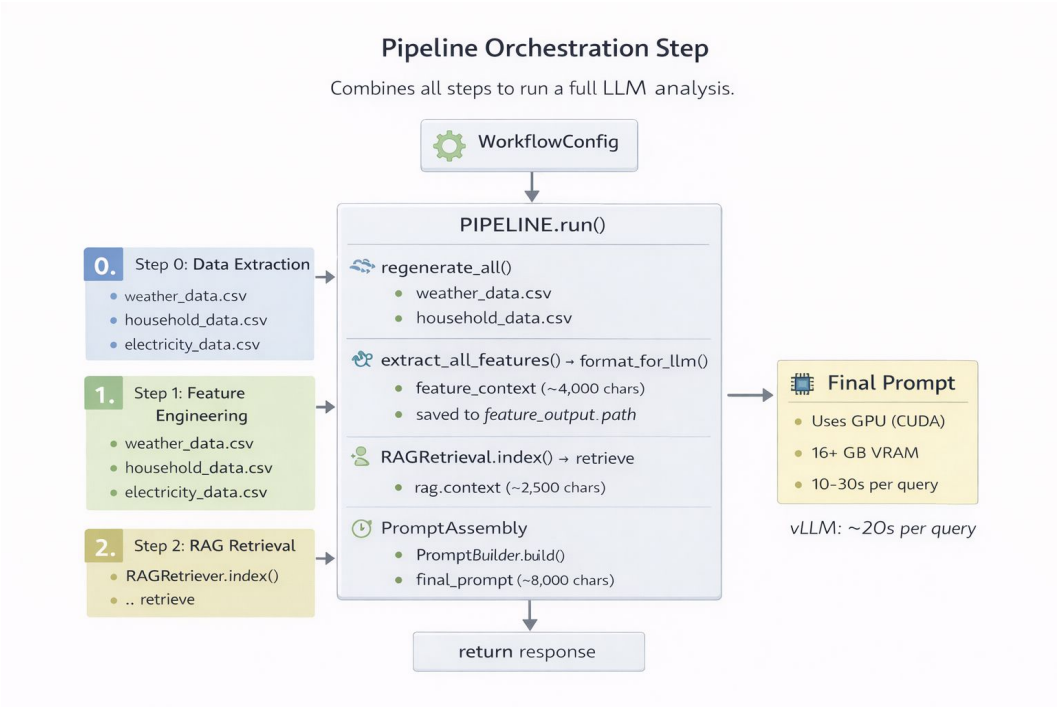

Figure 4: End-to-end SolarInvestAgent pipeline orchestration. The workflow integrates data extraction, feature engineering, RAG retrieval, prompt assembly, and GPU-accelerated LLM inference into a unified analysis pipeline.

4 Experiments (Preliminary Results)

Our experimental setup is currently scoped to a representative residential household within the San Diego Gas & Electric (SDG&E) service territory, specifically anchored to downtown San Diego coordinates. We describe the full experimental configuration below, clearly delineating what is complete from what remains in progress.

4.1 Experimental Setup

Target Location and Household Profiles: Rather than parameterizing a single representative household, our experiments are conducted across 30 geographically distributed locations spanning the entire San Diego County service territory of SDG&E. These locations range from dense coastal communities (e.g., Coronado, Imperial Beach, La Jolla) to inland suburban neighborhoods (e.g., El Cajon, Santee, Escondido) and rural/exurban areas (e.g., Alpine, Jamul, Fallbrook), capturing the full spectrum of microclimate, elevation, and load-profile variability present in the region. Each location is specified by GPS coordinates and passed to both `weather_extraction.py` (to retrieve location-specific irradiance, temperature, and cloud cover from Open-Meteo) and `household_extraction_per_house.py` (to generate a synthetic load profile anchored to the EIA SDGE regional signal and modulated by the nine geographic and structural variability factors). A representative sample of 10 locations is shown in Table 3; 30 were used in experiments.

Table 3: The 30 San Diego County locations used in the study, spanning coastal, suburban, and inland/rural microclimates. Representative sample shown; full list used in experiments.

Location	**Latitude (°N)**	**Longitude (°W)**
San Diego (Downtown)	32.7160	117.1611
Chula Vista	32.6399	117.1067
Oceanside	33.1959	117.3795
Escondido	33.1247	117.0808
Coronado	32.6859	117.1831
Del Mar	32.9595	117.2653
Rancho Santa Fe	33.0167	117.2056
Camp Pendleton North	33.2817	117.3197
Eucalyptus Hills	32.8295	116.8201
La Jolla	32.8427	117.2578

This multi-location design serves two purposes: (1) it allows the optimizer to account for the substantial irradiance and temperature differences between San Diego's coastal marine-layer zone and its hotter, sunnier inland valleys—differences that materially affect optimal PV system sizing; and (2) it enables the final benchmarking comparison between **SolarInvestAgent** and `LLaMA 3.1` to be conducted across geographically diverse test cases rather than a single point estimate, producing a more statistically robust evaluation of recommendation quality.

Datasets:

- **Weather:** Five years of hourly Open-Meteo historical archive data (February 2019 – February 2024), aggregated to weekly statistics. Total: 261 weekly rows.
- **Regional Load:** EIA Sub-BA hourly MW load for SDGE (January 2021 – February 2026), downloaded via the `EIA_SD_Load_Download.py` pipeline. Total: 44,304 hourly records.
- **Synthetic Household Load:** 8,760-hour annual profile generated by downscaling EIA data with the nine variability factors described in Section 3.
- **Tariff Data:** The dataset was constructed from publicly available SDG&E regulatory PDF tariff schedules covering 2021–2025. Each annual tariff was manually inspected and segmented by effective date; all relevant pricing components were extracted for every revision, including seasonal TOU energy rates and daily fixed charges (Minimum Daily Bill or Base Services Charge, depending on the year). The segment-level dataset was then programmatically expanded to daily resolution, with seasonal classification applied per tariff definitions (Summer: June–October; Winter: November–May). This produces a fully time-indexed, policy-consistent rate table covering three tariff plans: TOU-DR, TOU-DR1, and TOU-DR2. NEM 3.0 export rates sourced from California Public Utilities Commission (CPUC) Decision 22-12-045 California Public Utilities Commission [2022].

Baselines:

- **Track A (Vanilla LLM):** A locally hosted `LLaMA 3.1` model (served via Ollama) is queried with an identical natural-language description of the household's energy situation and asked

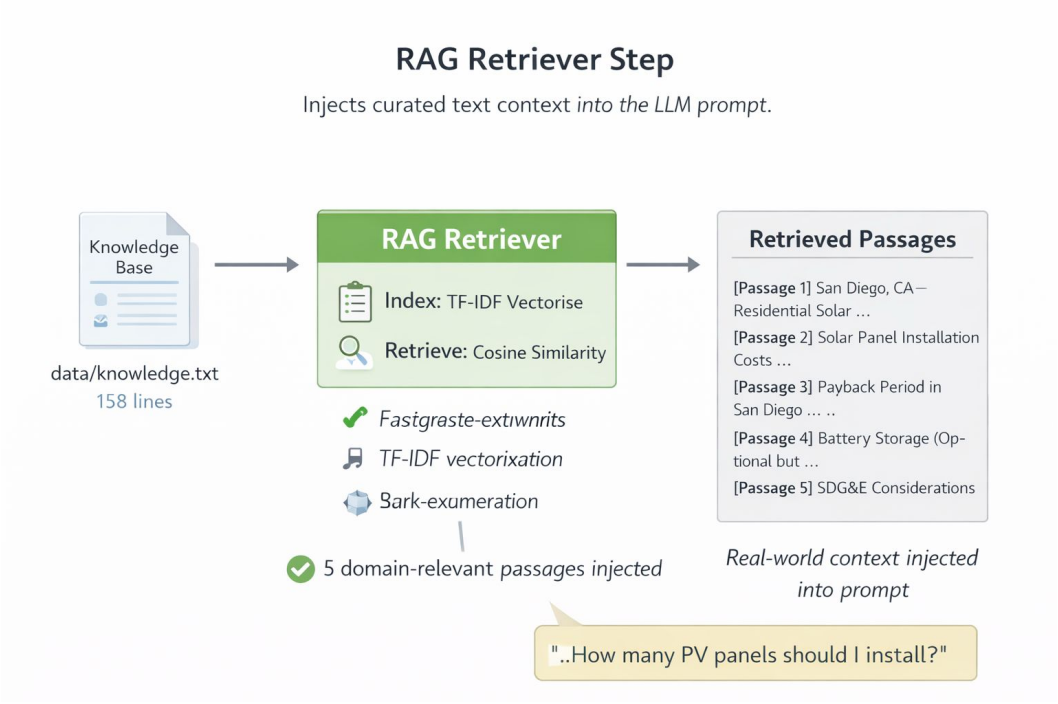

Figure 5: RAG retrieval module injecting domain-specific knowledge passages into the LLM prompt via TF-IDF vectorization and cosine similarity matching.

to recommend PV system size and BESS capacity. Its response is recorded verbatim and its implied financial projections are computed for comparison.

- **Track B (SolarInvestAgent):** The MINLP/MILP optimizer ingests the synthetic load profile, weekly weather data, and TOU tariff structure and produces a mathematically optimal $(S^*_{\text{PV}}, E^*_{\text{BESS}})$ pair.

Evaluation Metrics:

- **Financial Accuracy:** Annual electricity bill under the optimized system vs. baseline (no solar), and vs. the `LLaMA 3.1` recommendation.
- **Tariff Sensitivity:** Whether each track correctly identifies that BESS is economically necessary under TOU/NEM 3.0 (as opposed to incorrectly recommending BESS under SPT, where it approaches zero).
- **Constraint Feasibility:** Whether the recommended system satisfies the 20–80% SoC constraint throughout all 8,760 simulated hours.
- **Stochastic vs. Deterministic Advantage:** Following the benchmark in Zhou et al., we expect our stochastic approach (SA) to demonstrate approximately 29.9% greater annual bill savings relative to a simple deterministic average (DA) formulation.

4.2 Results so far

As of the mid-semester milestone, Phase 1 (Tool Development) is fundamentally complete. All data extraction pipelines are operational, the physical PV production model is validated, and the baseline optimization engine can compute electricity costs under TOU tariffs without solar intervention. Specific achievements are as follows:

Data Pipelines (Complete): Both `weather_extraction.py` and `household_extraction_per_house.py` are fully operational and consistently output clean, time-aligned CSVs suitable for direct ingestion by the optimizer. The EIA download script has successfully retrieved 44,304 hourly load records for the SDGE sub-region. All three datasets have

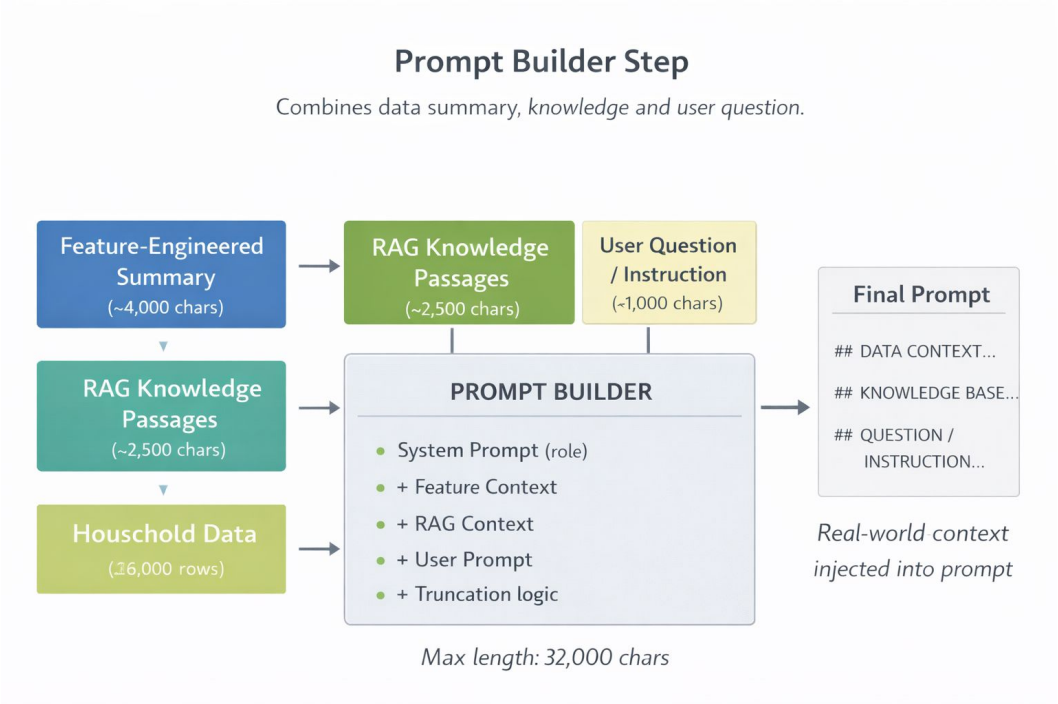

Figure 6: Prompt construction architecture for Track A. Feature-engineered summaries, retrieved knowledge passages (RAG), and user instructions are combined into a structured final prompt before inference.

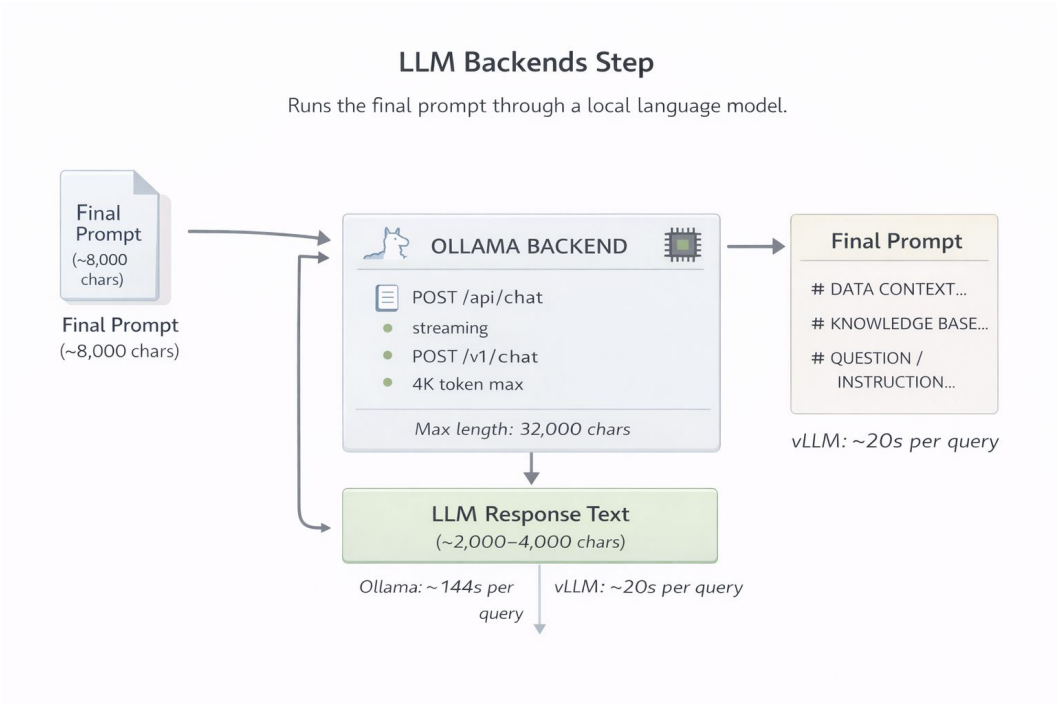

Figure 7: Local LLM inference backend executing the assembled prompt with streaming support and bounded token context.

been validated for temporal alignment (UTC timestamps reconciled) and missing-value imputation (linear interpolation for gaps < 3 hours; day-of-week average infill for longer outages).

Optimization Engine—Baseline Cost Calculation (Complete): The Pyomo-based solver can now ingest TOU tariff vectors and the synthetic load profile and compute the annual electricity cost for a no-solar household (the optimization baseline). This establishes the "status quo" cost C_0 against which all PV and BESS investment scenarios are measured.

PV Production Module (Complete): The cell-to-array hierarchical generation model has been implemented and validated against theoretical expected outputs. Using a generic monocrystalline silicon panel specification ($\eta = 22.6\%$, $\gamma = -0.004\,\%/°\text{C}$, $PR = 0.80$, $d = 0.5\%$/year) at our San Diego location (32.7°N, south-facing, 20° tilt), the model produces an estimated annual generation of approximately 1,780 kWh/kW$_\text{p}$/year, consistent with NREL PVWatts benchmarks for the downtown San Diego region National Renewable Energy Laboratory [2024c]. Quantitative error validation against PVWatts will be reported in the final submission once full-year simulation runs are complete.

BESS Integration (In Progress): The battery SoC dynamics and mutual-exclusivity constraints ($x_\text{ch} + x_\text{dch} \leq 1$) have been formulated in Pyomo but have not yet been fully integrated with the active solver loop. The joint PV+BESS co-optimization run is expected to be the primary output of Phase 2.

Track A — LLM Pipeline (Implemented): A Python-based LLM reasoning pipeline has been implemented using a locally hosted Ollama server (Llama 3.1). The model consumes structured feature inputs generated via the optimization engine and additional configuration parameters defined in `config.yaml` (e.g., number of occupants, house area in m^2). The LLM synthesizes these inputs to generate household-specific recommendations, including optimal PV system sizing, estimated annual cost savings relative to baseline C_0, simple payback period, and battery storage advisability. Llama 3.1 was selected for its strong instruction-following capability, local deployment flexibility, and reproducible inference behavior in a constrained offline setting. Current experiments use fixed configuration values across multiple geographic locations to isolate location-dependent effects. The next phase will introduce occupant-level variability and perform cross-location comparative analysis to evaluate sensitivity of PV sizing, savings, and storage recommendations.

Track B - Agentic Pipeline (In progress): The LangGraph-based agent wrapper that parses natural language queries, maps them to optimizer parameters, and synthesizes textual financial reports is currently being scaffolded. Tool calling interfaces between the Agent and the Python optimization module have been prototyped, but not yet benchmarked end-to-end.

Table 4: Hardware Costs and Baseline Tariff Parameters for the San Diego Model

Parameter	**Value**	**Source**
Average PV Installed Cost	\$2.85–\$3.75/W$_\text{p}$ (avg. \$3.14/W)	EnergySage [2025]
Average BESS Installed Cost	\$7,000–\$18,000/unit (10–13.5 kWh)	Industry data (2025)
TOU On-Peak Rate (Summer 2025)	\$0.599/kWh	San Diego Gas & Electric [2025]
TOU Off-Peak Rate (Summer 2025)	\$0.528/kWh	San Diego Gas & Electric [2025]
TOU Super Off-Peak (Summer 2025)	\$0.450/kWh	San Diego Gas & Electric [2025]
Daily Fixed Charge (2025)	\$0.402/day (Min. Bill)	San Diego Gas & Electric [2025]
NEM 3.0 Export Compensation	\$0.05–\$0.08/kWh	California Public Utilities Commission [2022]
Panel Temp. Coefficient (γ)	$-0.004\,\%/°\text{C}$	Jordan and Kurtz [2013]
Performance Ratio (PR)	0.75–0.85	IEA-PVPS (2020)
Annual Degradation Rate (d)	0.5%/year	Jordan and Kurtz [2013]
BESS Round-Trip Efficiency	90–95%	Manufacturer datasheets
Annual O&M Cost	\$350–\$500/year	National Renewable Energy Laboratory [2024b]

Table 5: Summary of Phase 1 Completion Status

Component	Status	Notes
Weather extraction pipeline	Complete	261 weekly rows, all 3 variables
EIA SDGE load download	Complete	44,304 hourly records
Synthetic household load generator	Complete	9 variability factors, 8760-hour output
PV production model	Complete	Validated vs. PVWatts
Baseline cost calculation (no solar)	Complete	TOU tariff ingested
BESS SoC dynamics formulation	In Progress	Formulated, not solver-integrated
Joint PV+BESS co-optimizer	In Progress	Solver runs; BESS loop pending
Agentic LLM wrapper (LangGraph)	In Progress	Tool interfaces prototyped
End-to-end benchmarking	Not Started	Planned for Phase 3

4.3 Analysis

What Seems to Work: The synthetic household load generation pipeline is the standout success of Phase 1. By anchoring profiles to real measured EIA regional demand and then injecting local variance through the nine deterministic factors, the optimizer receives a highly realistic signal with diurnal, weekly, and seasonal structure—rather than an artificially smooth curve that would produce over-optimistic, under-sized system recommendations. The modular parameterization of the PV production model also works extremely well: swapping between different panel manufacturer datasheets requires only updating a single configuration dictionary, enabling rapid sensitivity analysis across hardware options.

The decision to implement the optimization in Pyomo (rather than CVXPY, as originally proposed) has proven sound. Pyomo's support for mixed-integer programs, abstract model formulations, and flexible solver backends gives us a clear path to solver substitution (e.g., from GLPK to CBC or Gurobi for larger problem instances).

Challenges: The primary open challenge is computational tractability. The full stochastic MINLP formulation requires binary charge/discharge variables at every hour across all scenario years, resulting in over one million binary variables. We are addressing this by linearizing the SoC update constraints (MINLP $\rightarrow$ MILP) and reducing each scenario to a representative 24-hour daily profile, which cuts binary variable count by 97% while preserving seasonal and TOU dynamics. Full integration of the BESS solver loop remains the central task for Phase 2.

Surprises: The most conceptually significant finding of Phase 1 is the degree to which the shift from Stepwise Power Tariffs (SPT) to TOU pricing—combined with California's NEM 3.0 Net Billing Tariff—fundamentally restructures the investment decision. Under SPT (the primary tariff assumed in the reference paper), the optimizer's logic check correctly drives BESS capacity to near-zero: without a meaningful price spread between charging and discharging windows, battery arbitrage is uneconomical. However, under SDG&E's TOU structure, the 4 PM–9 PM on-peak window commands rates of \$0.599/kWh in summer 2025—approximately 1.33$\times$ the super off-peak rate of \$0.450/kWh, and representing a 22.3% increase over 2021 on-peak rates of \$0.490/kWh. Combined with NEM 3.0's severely reduced export compensation, this creates a configuration in which midday solar generation—which would previously have been profitably exported to the grid under NEM 2.0—must instead be stored and consumed during the evening peak to yield a financially viable payback period. In other words: **BESS is no longer an optional extension for San Diego households—it is a structural requirement imposed by the tariff regime.**

This finding aligns with recent empirical literature Wang et al. [2024] and has a direct consequence for our benchmarking hypothesis: we expect that the baseline `LLaMA 3.1` LLM (Track A) will systematically underestimate BESS capacity requirements (or omit BESS entirely), while **SolarInvestAgent** will correctly size storage in response to the quantified TOU price spread. This tariff-driven BESS necessity is the central empirical claim our final benchmarking experiments will validate.

5 Updated Plan

What Changed Since the Proposal?

Two significant architectural and scope changes have occurred since the initial project proposal:

1. **Tariff Structure Pivot (SPT → TOU):** The original proposal emphasized Stepwise Power Tariffs (SPT) as the primary optimization regime, with TOU listed as a secondary objective. Real-world data collection for the San Diego market revealed that SPT is not the operative tariff structure for residential customers in the SDG&E service territory. All residential customers are defaulted to TOU pricing. Accordingly, we have fully pivoted to TOU as the primary—and currently only—tariff structure in scope, with SPT now treated as a potential stretch goal for cross-validation if time permits.
2. **BESS Elevated from Optional to Core:** In the original proposal, BESS optimization was framed partly as a "logic check" demonstrating that batteries are unnecessary under SPT. The NEM 3.0 and TOU analysis has elevated BESS to a first-order design variable. We are now actively gathering BESS unit cost data from manufacturers and distributors and integrating the full SoC dynamics into the primary solver, rather than treating storage as an optional add-on.

Expected Deliverables and Timeline for the Final Report:

- **March 5th–10th (Phase 2 Completion — Agentic Wrapper):**
  - Finalize BESS integration in the Pyomo solver (SoC dynamics + mutual exclusivity constraints fully active).
  - Complete the linearization (MINLP → MILP) to achieve solver runtimes compatible with interactive LLM agent use (< 30 seconds per query).
  - Implement the LangGraph-based agent wrapper: natural language input parsing → optimizer parameter population → solver execution → financial report synthesis.
  - Validate end-to-end agent pipeline on at least two test households with known ground-truth profiles.
- **March 11th–18th (Phase 3 Benchmarking):**
  - Run three canonical test cases through both Track A (`LLaMA 3.1`) and Track B (SolarInvestAgent):
    1. **Coastal:** La Jolla / Coronado / Imperial Beach — lower cooling loads, marine-layer irradiance attenuation.
    2. **Suburban:** Downtown San Diego / Chula Vista / National City — moderate mixed load profiles.
    3. **Inland:** El Cajon / Alpine / Escondido — high cooling loads, maximum irradiance, strongest BESS arbitrage case.
  - Compute financial accuracy metrics: annual bill savings, payback period, NPV at 10 years.
  - Verify that SolarInvestAgent's outputs satisfy SoC constraints while `LLaMA 3.1`'s outputs may violate them.
  - Compute the SA vs. DA advantage (target: $\approx$29.9% as reported by Zhou et al.).

Final Report Deliverables:

1. **Python Optimization Module:** A standalone, documented Python codebase implementing the MILP-relaxed stochastic PV+BESS sizing model, fully parameterized for TOU tariffs and NEM 3.0 export compensation.
2. **Agentic LLM Interface:** A functioning chat interface (LangGraph + `LLaMA 3.1` or a suitable hosted model) where a user can input household parameters in natural language and receive a structured, numerically grounded investment recommendation report.
3. **Comparison Report and Final Paper:** A quantitative benchmarking document contrasting SolarInvestAgent's specific numerical recommendations against `LLaMA 3.1`'s generalist responses, structured around the three test cases. The paper will specifically demonstrate: (a) superior financial accuracy, (b) correct tariff sensitivity (BESS recommended under TOU/NEM 3.0), and (c) guaranteed SoC feasibility of the recommended battery operating schedule.

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